

Final Exam Review

Experiment

- In certain cases you want to make a statement about a particular UI design choice

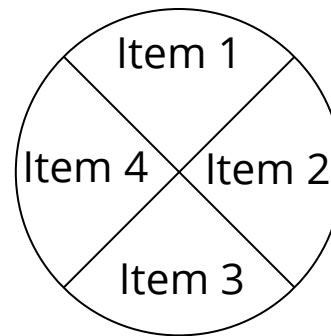
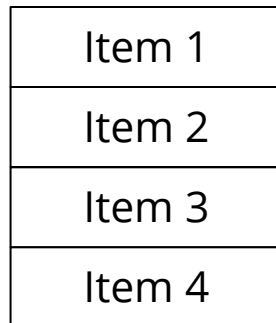
BUTTON

BUTTON

BUTTON

- “I want to know whether the size of buttons contribute to how quickly users click on them”

Or



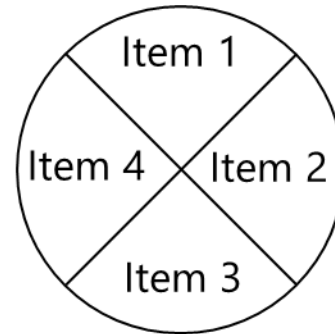
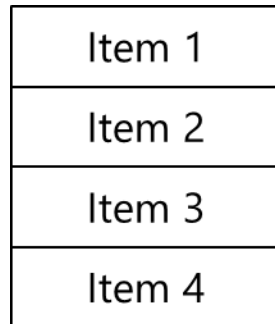
- “I want to know whether a menu designed in a circular shape (pie menu) is more effective (e.g., time, error) than a regular menu”

Example

- How do we test this question?
- Need to come up with
 - a hypothesis
 - a set of variables we are going to manipulate
 - a set of variables we are going to measure
 - a task
 - a set of trials
 - ...

Hypothesis Testing

- Null hypothesis (H_0)
 - There exists no relationship between manipulating the independent variables and the resultant changes in the dependent variables
- Example:
 - “There is no difference in selection speed between pie-menus and vertical context menus”



- In testing hypothesis, we are seeking to reject the null hypothesis

Hypothesis

- The alternative hypothesis (H_a) claims " H_0 is false"
- H_0 (null hypothesis): There is no difference in the number of cavities in children and teenagers using *crest* and *no-teeth* toothpaste when brushing daily over a one month period
- H_a (alternative hypothesis): children and teenagers using crest toothpaste have fewer cavities than those who use *no-teeth* toothpaste



Independent Variables

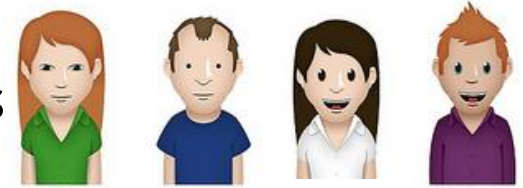
- Hypothesis: There is no difference in user performance (time and error rate) when selecting a single item from a pop-up or a pull down menu of length 3, 6, 9 or 12 items, regardless of the subject's previous expertise in using a mouse
- What are the independent variables? What are the levels?
 - Menu type: pop-up or pull-down
 - Menu length: 3, 6, 9, 12
 - Subject type: expert or novice

Example

- There is no difference in the number of cavities in children and teenagers using glow-right and no-teeth toothpaste when brushing daily over a one month period in toothpaste experiment
- Dependent variable: number of cavities
- Other things we could have measured
 - preference

Study Design Decisions

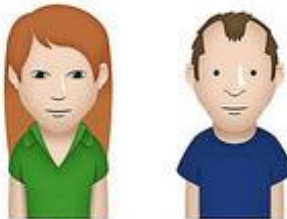
- How will participants be assigned to conditions
- Two options:
 - Between-subjects
 - Within-subjects



Condition 1



Between-subjects



Within-subjects

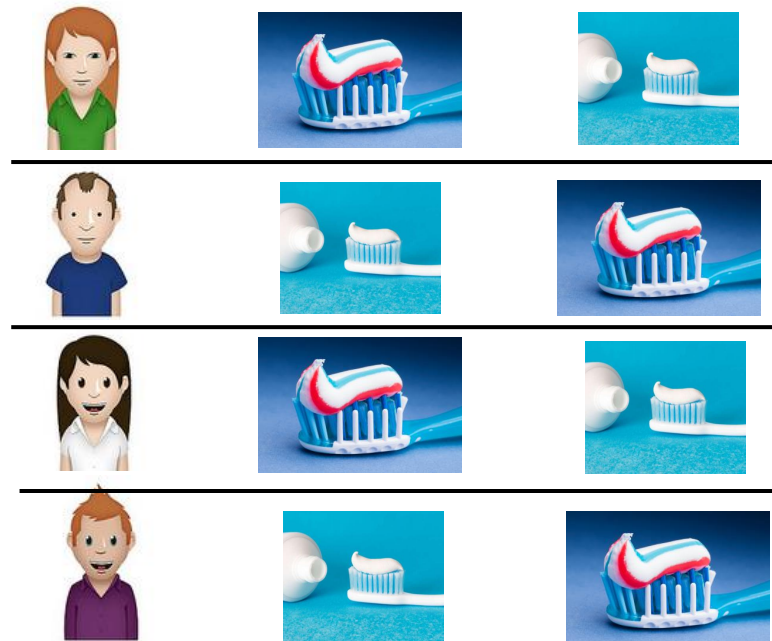


Condition 2



Order Effects, Counterbalancing

- Only relevant for within-subjects factors
- Order effects offset by counterbalancing:
 - Participants divided into groups
 - Test conditions are administered in a different order to each group
- Order of administering test conditions uses a Latin square



Heuristics

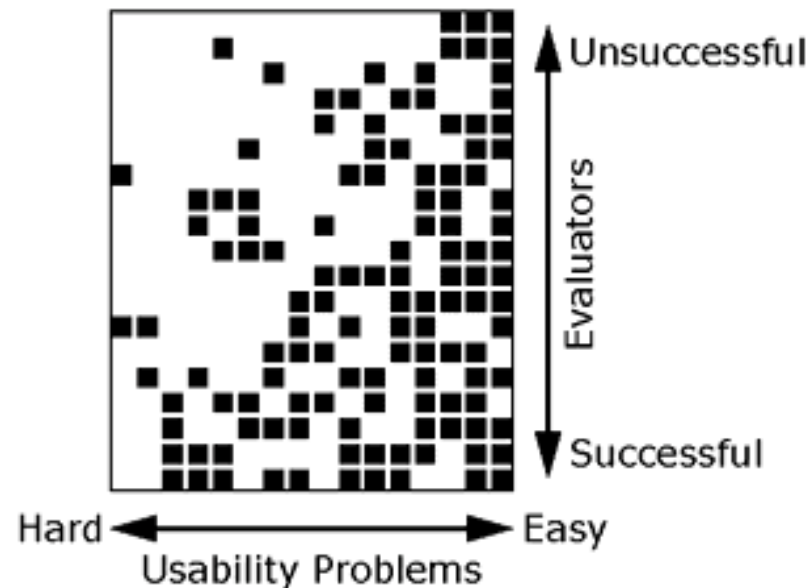
- Heuristics are collection of rules and guidelines
- Help to identify design problems
- Traditionally quite large collections
 - Brown 1988: 302 rules
 - Smith & Mosier 1986: 944 rules
- Nowadays lighter methods have become used
 - [Nielsen](#): 10 rules
 - [Schneiderman](#): 8 rules



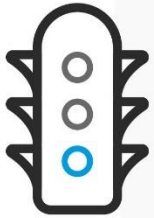
Jakob Nielsen

Heuristic Evaluation

- Different people identify different errors
- 19 evaluators were used to find 16 usability problems in a voice response system allowing customers access to their bank accounts (Image below - Nielsen 1992)
- Some usability problems are so easy to find that they are found by almost everybody
- Some problems that are found by very few evaluators



10 Usability Heuristics



**Visibility of
System Status**



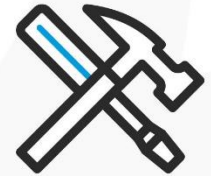
**Match Between System
& the Real World**



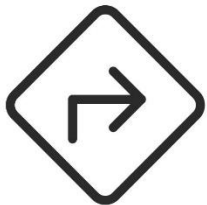
**User Control
& Freedom**



Consistency & Standards



Error Prevention



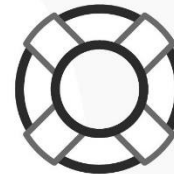
**Recognition Rather
than Recall**



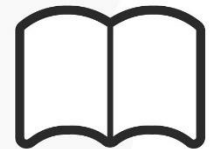
**Flexibility &
Efficiency of Use**



**Aesthetic &
Minimalist Design**



**Help Users Recognize, Diagnose
& Recover from Errors**



**Help &
Documentation**

Dimensions to Classify Computer Input

- Sensing Method
 - mechanical (e.g. switch)
 - motion (e.g. accelerometer, gyroscope)
 - contact (e.g. capacitive touch, pressure sensor)
 - signal processing (e.g. computer vision, audio)
- Continuous vs. Discrete
- Degrees of Freedom (DOF)
 - number of independent motions that are allowed to the body



Dimensions to Classify Computer Input

- iPod wheel: 1 DOF contact sensing continuous, 1 DOF discrete mechanical buttons (assume one button at a time)
- Mouse: 2 DOF computer vision (used to be mechanical), 1 DOF mechanical wheel, 1 DOF discrete mechanical buttons
- Wii: 6 DOF continuous translation and rotation, 1 DOF discrete mechanical buttons
- keyboard: ~1 DOF discrete mechanical keys



Perceived Qwerty Problems

- Common combinations require awkward finger motions. (eg: tr)
- Common combinations require a jump over home row. (eg: br)
- Common combinations are typed with one hand. (e.g. was, were)
- Most typing with the left hand
- About 16% of typing uses lower row, 52% top row, 32% home row



Dvorak Typing Movement Optimizations

- Letters should be typed by alternating between hands
- For maximum speed and efficiency, the most common letters and digraphs should be the easiest to type. Thus, about 70% of keyboard strokes are on home row.
- The least common letters should be on the bottom row, which is the hardest row to reach.
- The right hand should do more of the typing, because most people are right-handed.



Predictive Text

- Use language characteristics to predict input
 - Given characters typed so far, what are the most likely intended words?
 - Given words typed so far, what is the most likely word to follow?
- A variation is used for T9, nine-key text entry
 - Given an ambiguous set of characters, what is the most likely word
- Possible Problems
 - “collisions” between common words
 - entering words not in dictionary difficult
 - hard to focus on typing and monitoring prediction



Absolute vs. Relative Position

- Absolute position is a direct mapping (i.e., 1-to-1) of input device position to a display input position



Indirect



Direct

- Relative position maps changes in input device position to changes in display input position



- To make relative position work, you need a “clutch”

Direct vs. Indirect Input

- Indirect:
 - The position of the cursor is controlled by some external devices

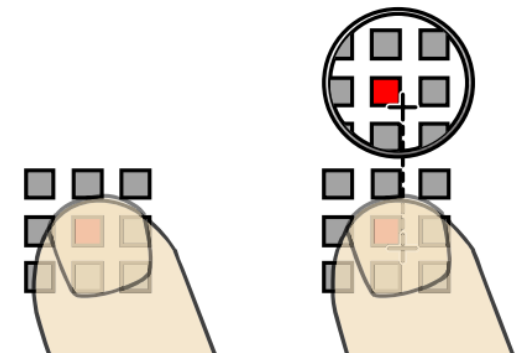
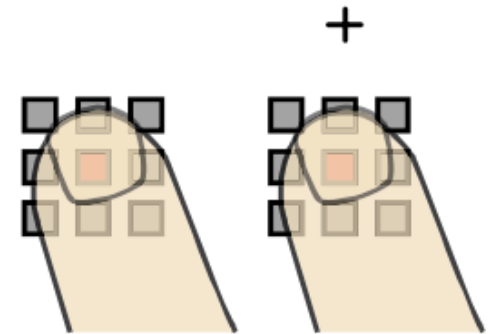
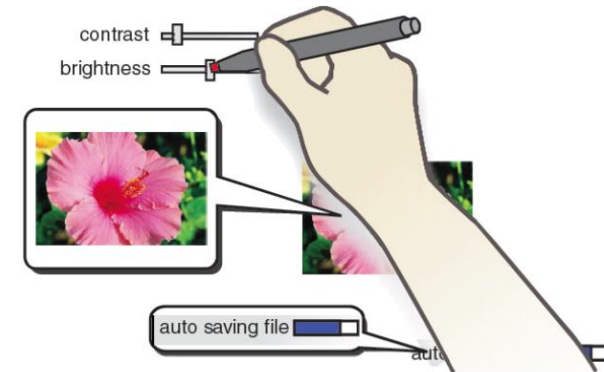


- Direct:
 - the position of the cursor is controlled by direct contact with the screen.



Occlusion-Aware Interfaces

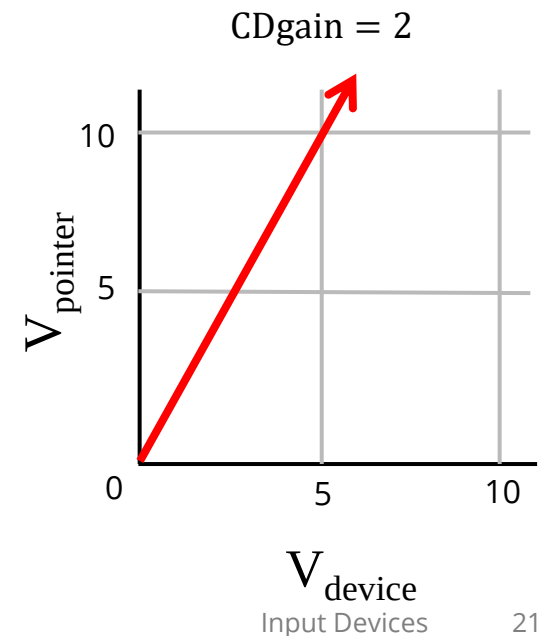
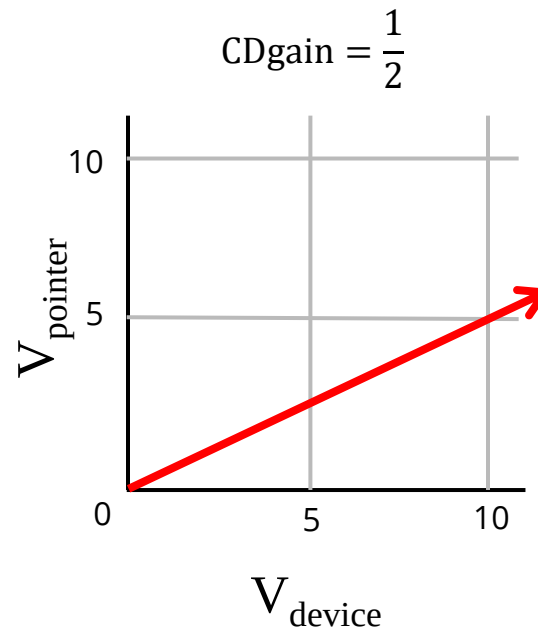
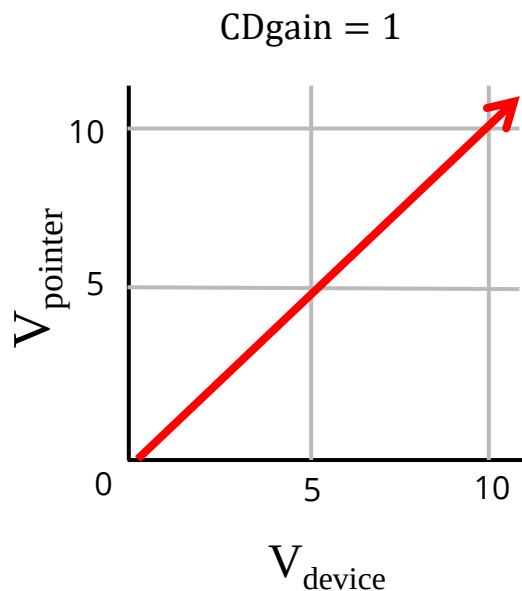
- Occlusion-Aware Viewer technique: [Vogel & Balakrishnan](#), 2010
 - Displays otherwise missed previews and status messages in a non-occluded area using a bubble-like callout.
- Offset Cursor. B. Shneiderman, 1991
 - Create an offset between pointer and finger
 - Cannot aim for the actual target anymore
 - Unreachable areas (e.g., bottom of the screen)
- Shift. [Vogel and Baudisch](#). 2007.
 - Shift creates a callout showing a copy of the occluded screen area and places it in a non-occluded location



Control-Display Gain (CD Gain)

- Maps the movement of the pointing device to the movement of the display pointer
 - Ratio of **display pointer** movement to **device control** movement

$$\text{CDgain} = V_{\text{pointer}} / V_{\text{device}}$$



Properties of Positional Input Devices

- Absolute vs. Relative Positioning
 - touchscreen = absolute
 - mouse = relative
- Direct vs. Indirect Contact
 - direct = touchscreen
 - indirect = mouse
- Force vs. Displacement Sensing
 - (most) joysticks = force
 - mouse = displacement
- Position vs. Rate Control
 - (most) joysticks = rate
 - mouse = position
- DOF (Dimensions) Sensed
 - 2 = mouse, 6 = Wiimote



KLM Example (Only Physical Operators)

- Use KLM to compare the performance time of three different date entry widgets. (assume: **hand already on mouse**, **40 WPM typist**)

- One text field

“MM/DD/YYYY”



PB H KKKKKKKKKK = 4.6s

- Steps:

1. Move the mouse cursor to the box/edittext (P)
2. Click on the box/edittext (B)
3. Move the hand from the mouse to the keyboard (H)
4. Type 10 characters, including “/” (10K)

Op	Time
K	0.3
P	1.1
B	0.1
H	0.4
M	1.2

KLM Example (Only Physical Operators)

- Use KLM to compare the performance time of three different date entry widgets. (assume: **hand already on mouse**, **40 WPM typist**)

- Three Dropdowns



PBPB PBPB PBPB = 7.2s

- Steps:
 1. Move the mouse cursor to the first dropdown (P)
 2. Click on the first dropdown list to see the items (B)
 3. Move the cursor to select an item (P)
 4. Select an item (B)
- Repeat Step 1-4 for the second and third dropdown lists

Op	Time
K	0.3
P	1.1
B	0.1
H	0.4
M	1.2

KLM Example (Only Physical Operators)

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- One text field

"MM/DD/YYYY"



PB H KKKKKKKKKKK = 4.6s

- Three Dropdowns



PBPB PBPB PBPB = 7.2s

- Three text fields



With tab: PB H KK K KK K KKKK= 4.6s

Without tab: PB H KK HPB H KK HPB H KKKK= 8.0s

Op	Time
K	0.3
P	1.1
B	0.1
H	0.4
M	1.2

Including Mental Operators (M)

- People need to think about something before doing it
 - identify when people have to stop and think: **M**
 - difference between actions using *cognitive conscious* and *cognitive unconscious*
- Insert an **M** operation when people have to:
 - initiate a task
 - retrieve a chunk from memory
 - find something on the display (e.g. point to something)
 - think of a task parameter
 - verify that a specification/action is correct (e.g. display changes)

KLM Example (Including Mental Operators)

- Use KLM to compare the performance time of three different date entry widgets. (assume: hand already on mouse, 40 WPM typist)

- One text field

"MM/DD/YYYY"



MPB H KKKKKKKKKKK = 5.8s

- Three Dropdowns



MPBMPB PBMPB PBMPB = 12s

- Three text fields



With tab: MPB H KK K KK K KKKK= 5.8s

Without tab: MPB H KK HPB H KK HPB H KKKK= 9.2s

Op	Time
K	0.3
P	1.1
B	0.1
H	0.4
M	1.2

Fitts' Law

- Fitts' Law: a predictive model for pointing time considering device, distance, and target size
 - published 1954
 - based on rapid, aimed movements
 - works for many kinds of pointing “devices”:
finger, pen, mouse, joystick, foot, ..
- Paul Fitts
 - Psychologist at Ohio State University
 - Early advocate of user-centred design
(in terms of matching system to human capabilities)



Index of difficulty

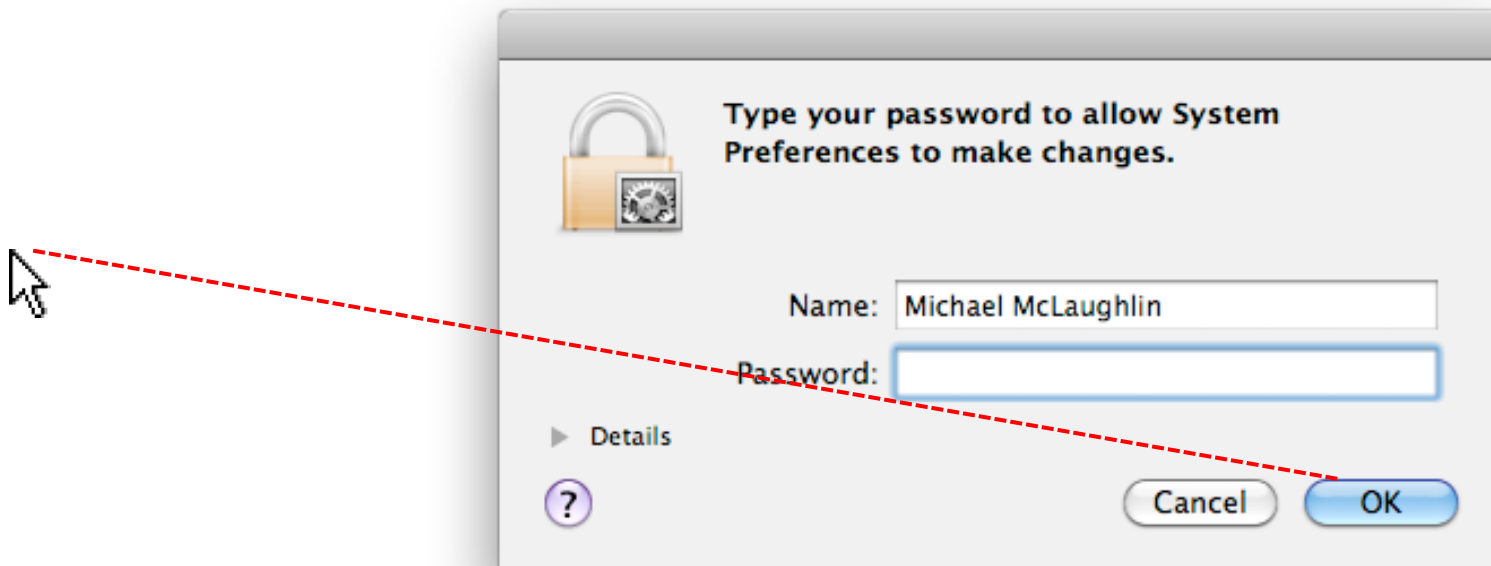
$$MT = a + b \underbrace{\log_2 \left(1 + \frac{D}{W} \right)}$$

- Index of Difficulty, ID = $\log_2 \left(1 + \frac{D}{W} \right)$
 - $MT = a + b \cdot ID$
 - ID how difficult a task is independent from the input device

- Units:
 - a is measured in seconds
 - b is measured in seconds per bit
 - Index of Difficulty (ID) is described in bits
 - a and b are constants determined through linear regression

Fitts' Law Example

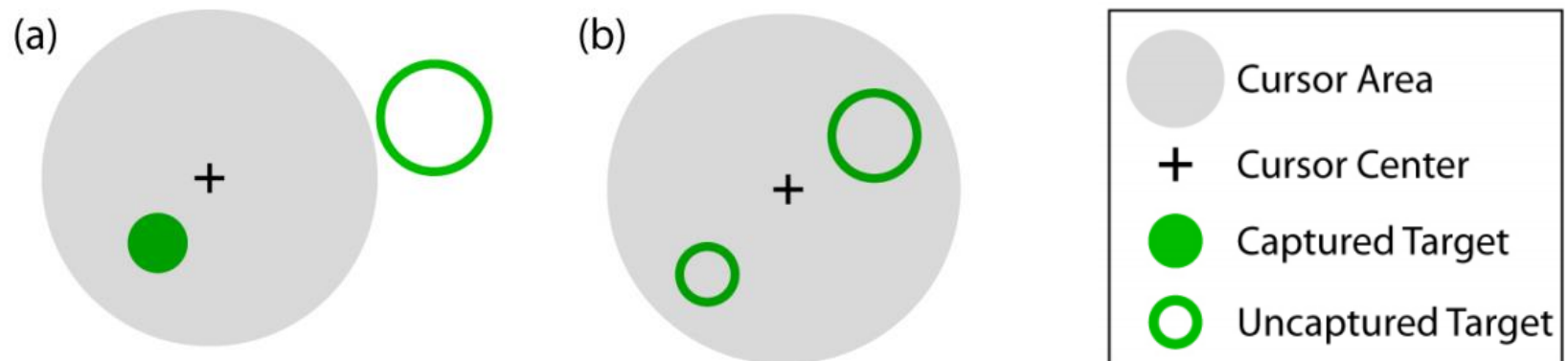
- Using a mouse to point ($a = -107$ and $b = 223$), what is the movement time to click on a 80 pixel by 32 pixel Cancel button located 400 pixels away?



$$MT = a + b \log_2 \left(\frac{D}{\min(W, H)} + 1 \right)$$

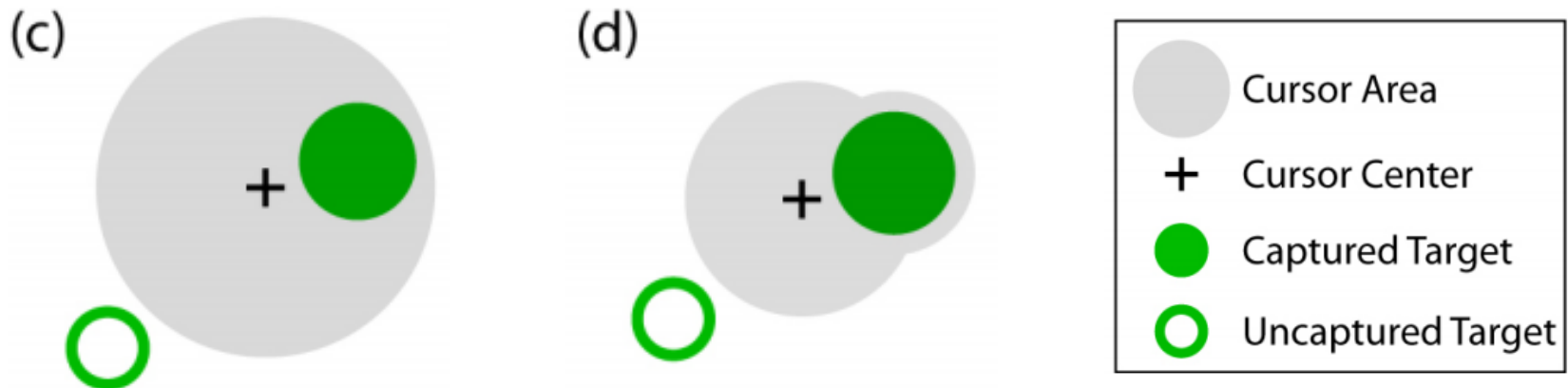
Area Cursor

- A standard point cursor commonly has a single point of activation or hotspot
- Area cursors have larger hotspots defined by the boundary of the cursor (Figure a)
- Problems arise, however, when the area cursor encompasses more than one target
- Making it difficult to isolate the intended target (Figure b)



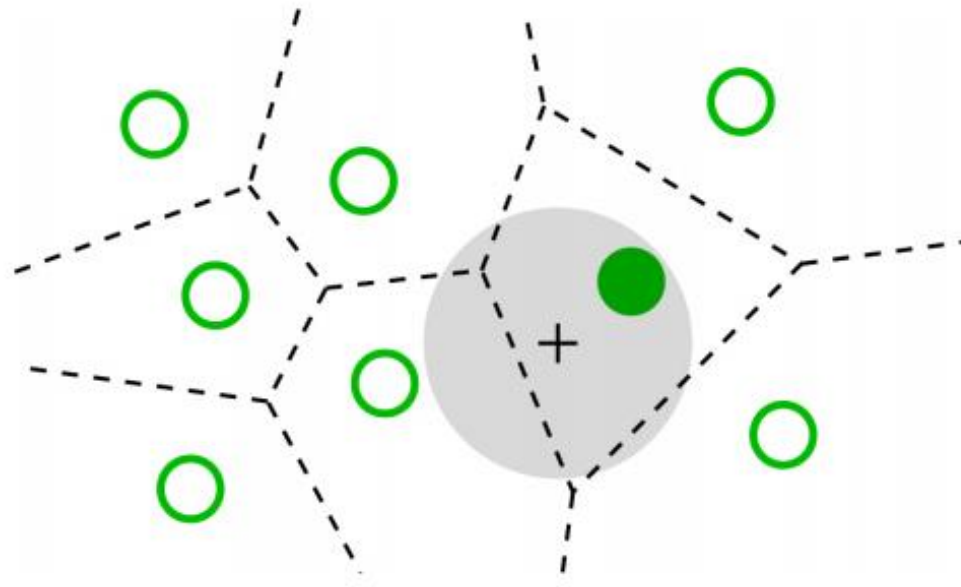
Bubble Cursor

- The bubble cursor solves this problem of the area cursor by dynamically updating its size based on the proximity of surrounding targets, such that there is always exactly one target inside the hotspot (Figure c, d).
- The bubble cursor morphs to encompass a target when the basic circular cursor cannot completely do so without intersecting a neighboring target (Figure d).



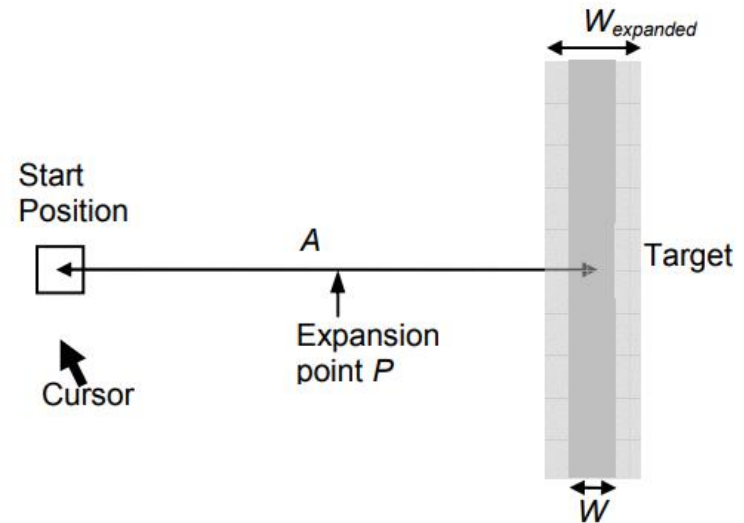
Bubble Cursor

- Fitts Law: decreasing D and increasing W
- Using the bubble cursor essentially divides up the total space in which all targets reside into regions
- There is exactly one target inside each region, and that target is the closest target to any point within that region. In mathematical terms, this is referred to as a **Voronoi diagram**



Acquisition of Expanding Targets

- The size of the interface widget or viewing region dynamically changes to provide the user with a larger target area
 - What happens when the target size changes after the user has already begun movement towards that target
-
- In the base case, the target had a width of W .
 - In the expanded cases, the target began with a width W but expanded to W_{expanded} when the cursor moved past a specified expansion point P .



Acquisition of Expanding Targets

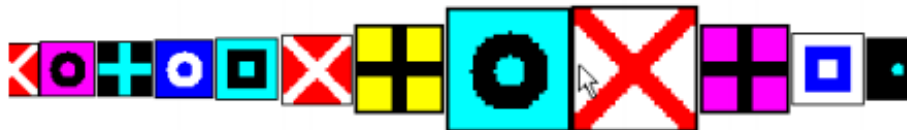
- Results indicate that such tasks can indeed be accurately modeled by Fitts' law.
- User performance is consistently aided by the target expansion, even when the target only expands after 90% of the distance towards the target has been traversed.



(a)



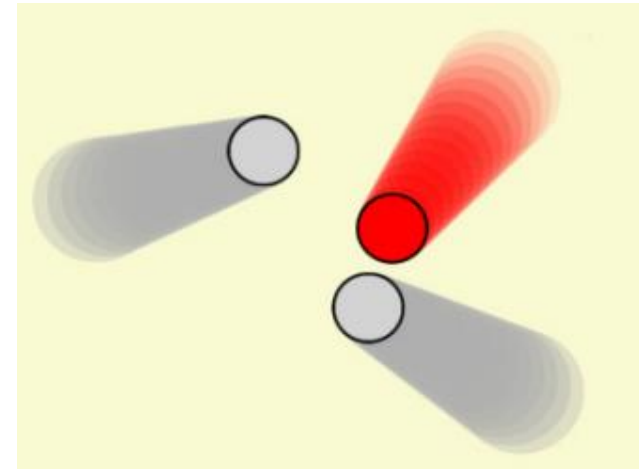
(b)



(c)

Comet and Target Ghost

- Comet, a technique that enhances targets based on their speed and direction
- With Comet the target's activation size is increased with an appended tail, analogous to comets that are seen in the sky
- Compared with bubble cursor for 1D (movement along x-axis) and 2D selection tasks
- In a 1D selection task, Comet outperformed the Area and Bubble cursors
- In a 2D selection task, Comet outperformed the Bubble, but only in conditions without the ghost



Khalad Hasan, Tovi Grossman, and Pourang Irani. 2011. Comet and target ghost: techniques for selecting moving targets. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (CHI '11). ACM, 839–848.

(Accot-Zhai) Steering Law

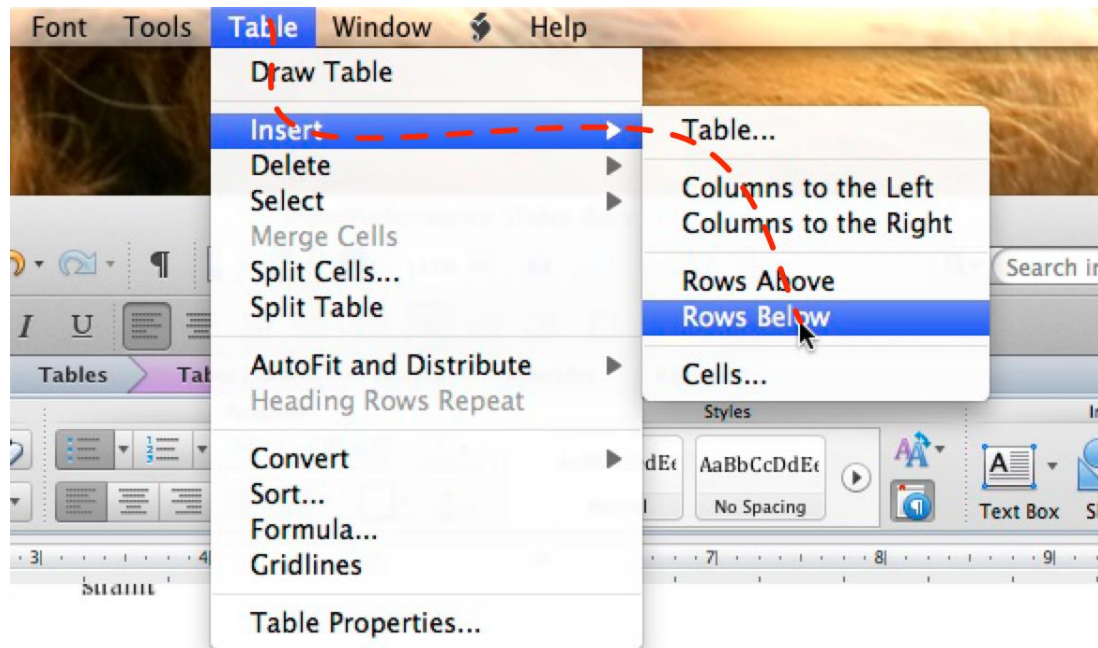
- Accot, J., & Zhai, S. (1997). Beyond Fitts' law: models for trajectory-based HCI tasks. CHI'97
- Tunnel traveling task is not related to the logarithm of D/W but to D/W

$$MT = a + b \frac{D}{W} \quad ID = \frac{D}{W}$$

- The movement time (MT) to acquire a target through a tunnel is a function of the length (D) and width (W) of the tunnel. It depends on the input device.
 - MT: movement time
 - a and b: constants dependent on the pointing system
 - D: distance, i.e. length of the tunnel
 - W: width of the tunnel

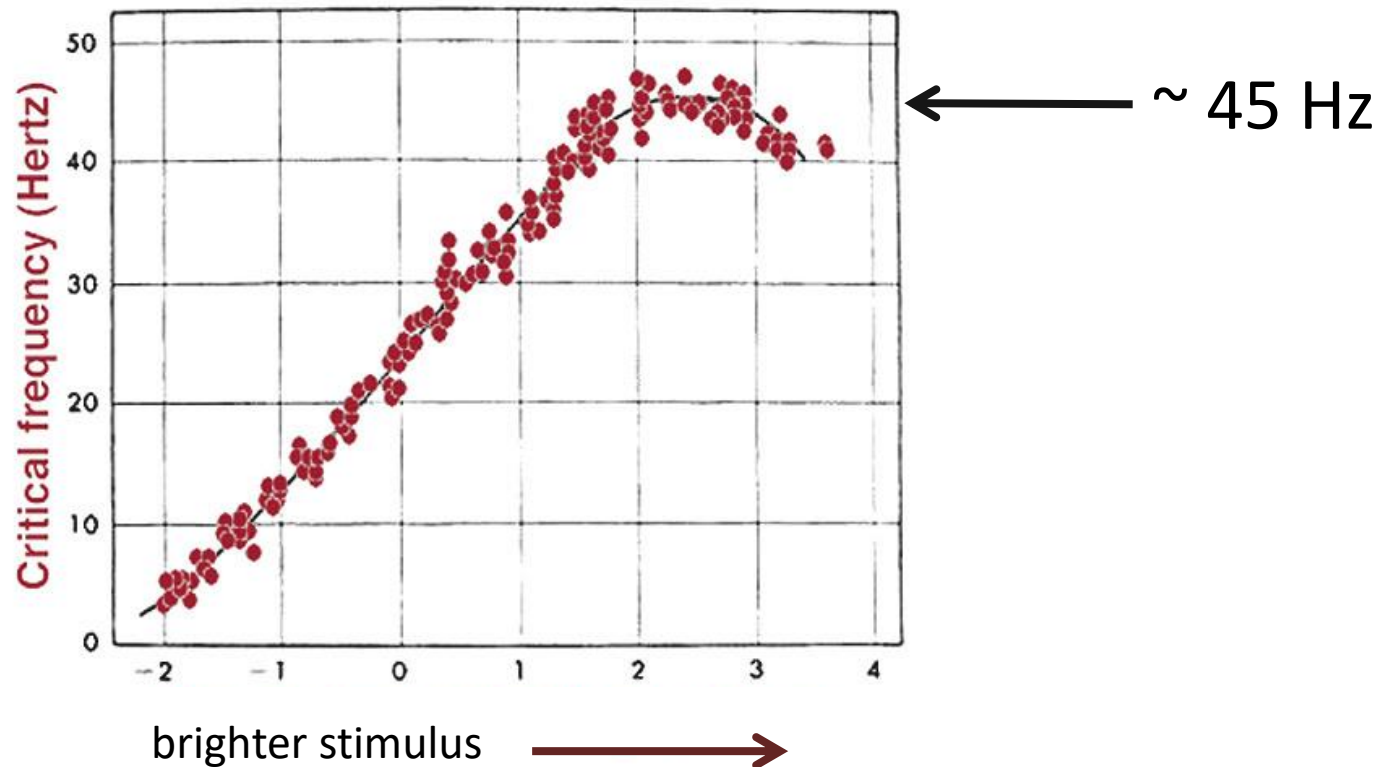
Hierarchical Menus

- Sum the parts of the path:
 - Wide path (but short stopping distance)
 - Narrow path (but wide stopping distance)
 - Wide path (with short stopping distance)



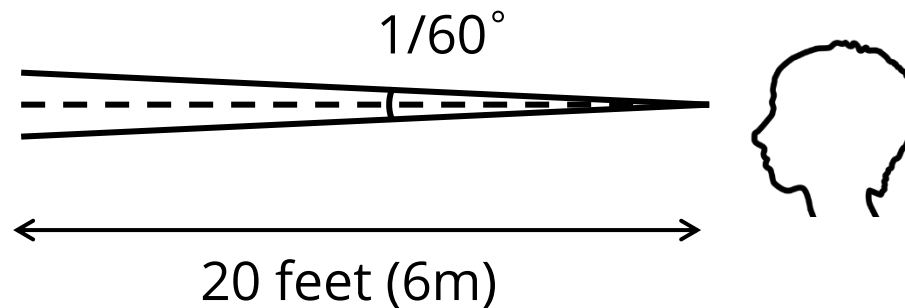
Temporal Resolution: Flicker

- Critical Flicker Frequency (CFF)
 - when perception of intermittent light source changes from flickering to continuous light
 - dependent on brightness of stimulus, wavelength, others ...



Spatial Resolution: Visual Acuity

- spatial resolution of visual processing system
 - 20/20 (6/6) vision: separate lines 1 arc minute ($1/60^\circ$) apart at 20 feet (6 m)



20/20 → Normal vision

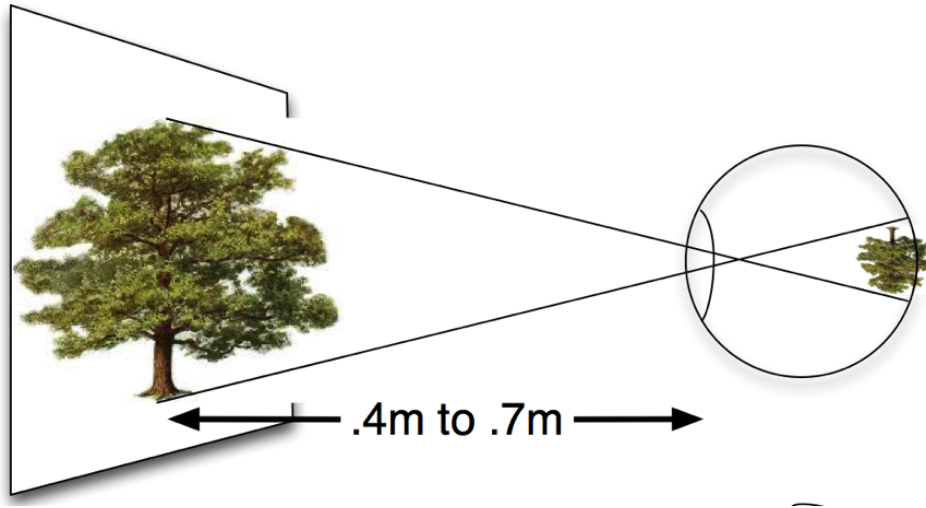
20/15 → A test subject sees at 20 feet that a person with 20/20 vision can see at 15 feet.

20/40 → A test subject sees at 20 feet what a person with normal vision sees at 40 feet

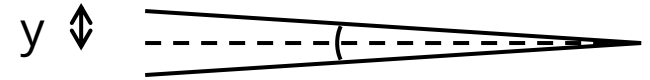
E	1	20/200
F P	2	20/100
T O Z	3	20/70
L P E D	4	20/50
P E C F D	5	20/40
E D F C Z P	6	20/30
F E L O P Z D	7	20/25
D E F P O T E C	8	20/20
L E F O D P C T	9	
F D P L T C E O	10	
P E Z O L C F T D	11	

Spatial Resolution Implications

- Best pixel density for displays?
 - density is ppcm (pixels per cm)

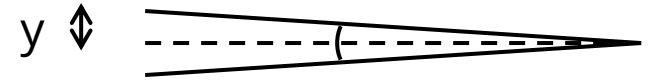
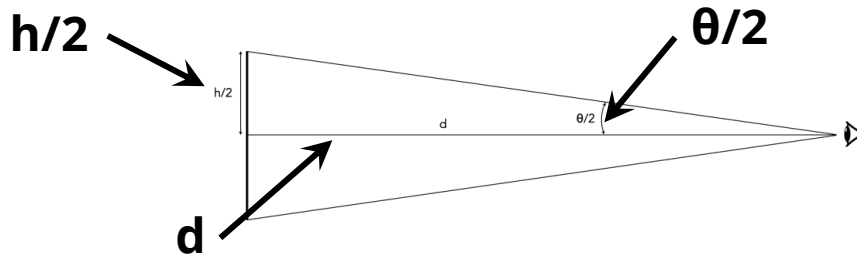


$$\theta = 1/60^\circ$$



Spatial Resolution Implications

$$\theta = 1/60^\circ$$



$$\tan(\theta/2) = (h/2)/d$$

$$h/2 = d \tan(\theta/2)$$

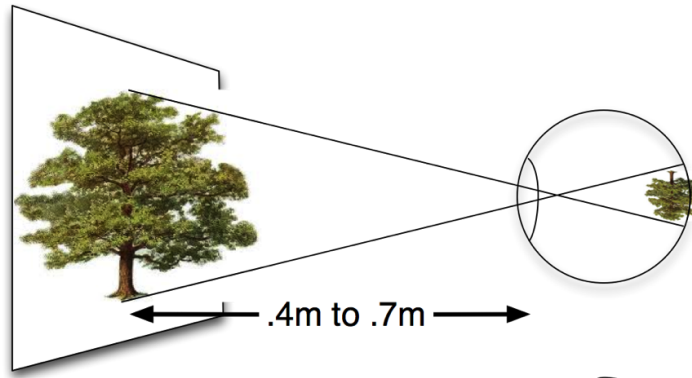
$$h = 2d \times \tan(\theta/2)$$

1 arcminute, i.e., $1/60^\circ$, is equivalent
to 0.000290888 radians

$$\begin{aligned} h &= 2d \times \tan(0.000290888/2) \\ &= 2d \times 0.000145444 \end{aligned}$$

Spatial Resolution Implications

- Best pixel density for displays?
 - density is ppcm (pixels per cm)



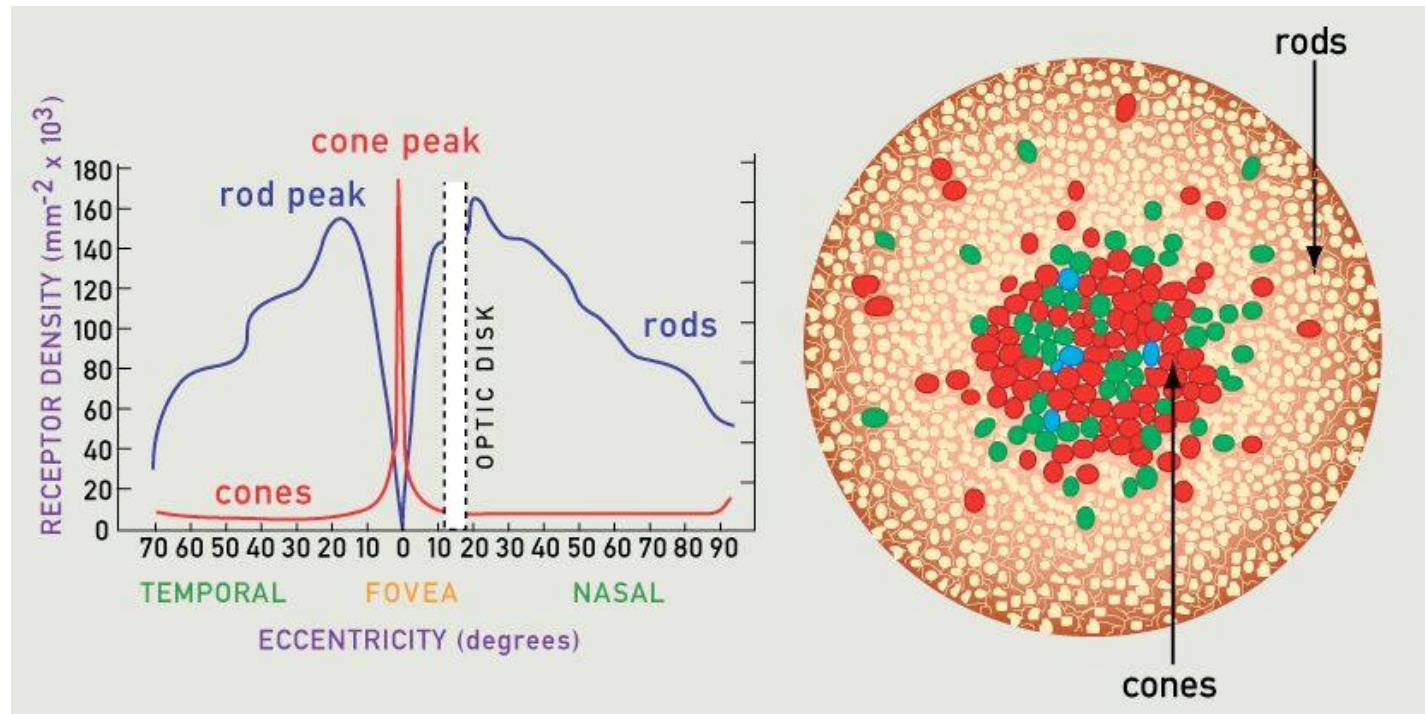
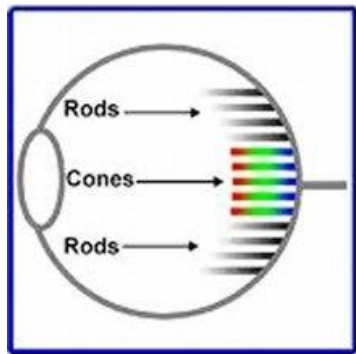
$$h = 2 \times 0.4 \times 0.000145444 = 0.116\text{mm (at } 0.4\text{m)}$$

$$h = 2 \times 0.7 \times 0.000145444 = 0.203\text{mm (at } 0.7\text{m)}$$

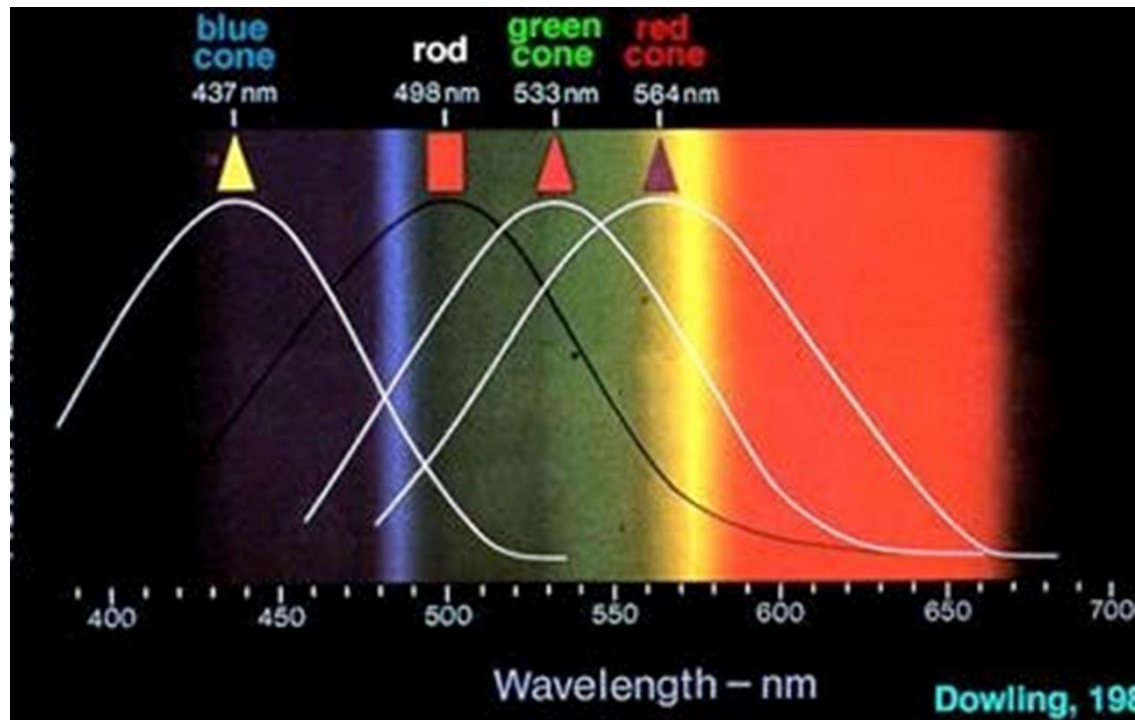
We can see individual pixels larger than about 0.116 to 0.203 mm

Colour Perception

- Two different light sensors in human eye
 - **Cones** perceive colour (focus, ~7 million)
 - **Rods** distinguish light from dark (peripheral vision, ~120 million)
- Cones and rods not evenly distributed across the retina
 - spatial resolution of visual field drops significantly at edges



3 Types of Cones means Trichromatic Vision

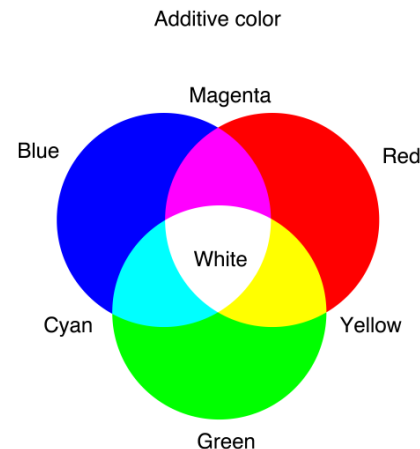


- Blue, green, and red cones
- Each is sensitive to a different band in the spectrum
- Few blue cones (approx: 64% red, 32% green, 4% blue)
- Harder to notice blues than reds

Colour Models

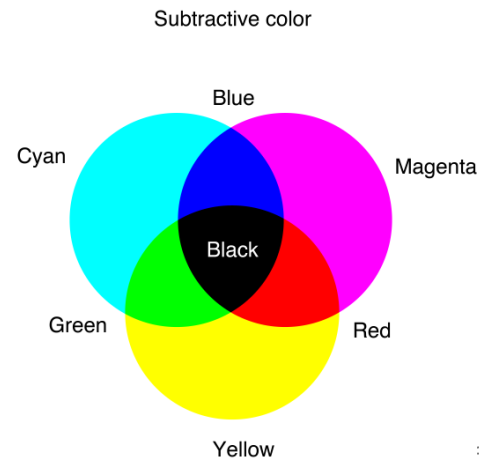
▪ Additive

- coloured light is added to produce white
- TVs, computer monitors, and other electronics use additive color
- Every pixel starts as black, and take on colors that are expressed as percentage values of red, green, and blue (RGB)



▪ Subtractive

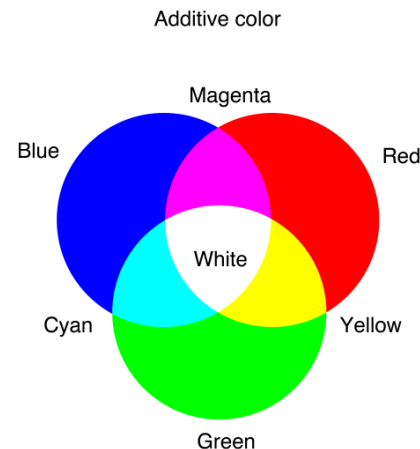
- coloured light is absorbed to produce black
- Subtractive colors begin as white.
- CMY/CMYK – Photos, magazines, and any printed material



Colour Models

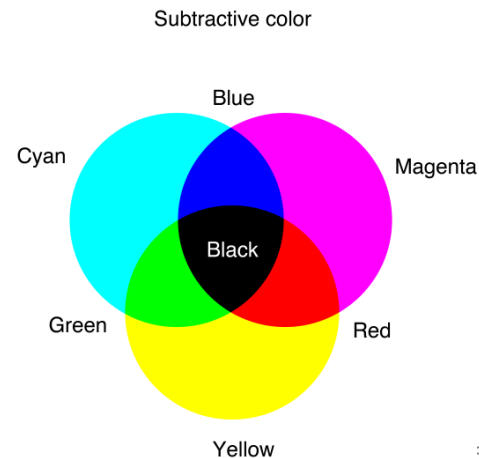
▪ Additive

- coloured light is added to produce white
- TVs, computer monitors, and other electronics use additive color
- Every pixel starts as black, and take on colors that are expressed as percentage values of red, green, and blue (RGB)



▪ Subtractive

- coloured light is absorbed to produce black
- Subtractive colors begin as white.
- CMY/CMYK – Photos, magazines, and any printed material



Gestalt Principles

- Theories of visual perception that describe how people tend to organize visual elements into groups or unified wholes, when certain principles are applied.
- Clues about how the brain groups raw visual input

Proximity

Similarity

Continuity

Closure

Connectedness

Figure and Ground

Hierarchy

- One kind of hierarchy is visual: use visual variables to make the more **important elements pop out**
- The structure of information on a page that communicates order, emphasis, and relationship.
- Create the hierarchy to support the intended reading sequence

Guiding the eye on the page so that it attends to different design elements in the order of their importance

Typography

- The practice of arranging written subject matter
- Difference between “Typeface” and “Font”
- Typeface refers to a family of fonts sharing the same name, like “Arial” or “Georgia.”
- A font is a choice of typeface and size and style, like roman, italic, oblique, boldface, etc



The quick brown fox

The quick brown fox

The quick brown fox

The quick brown fox

Times New Roman

The quick brown fox

The quick brown fox

The quick brown fox

The quick brown fox

Open Sans

Type Styles

- Style (Sans-Serif, Serif, Display, etc)
 - Serifs are the little handles on some fonts, a decorative stroke that finishes off the end of a letters stem
 - Sans-serifs: literally without the serifs



VOGUE

J.P.Morgan

T-Mobile



SONY



hulu

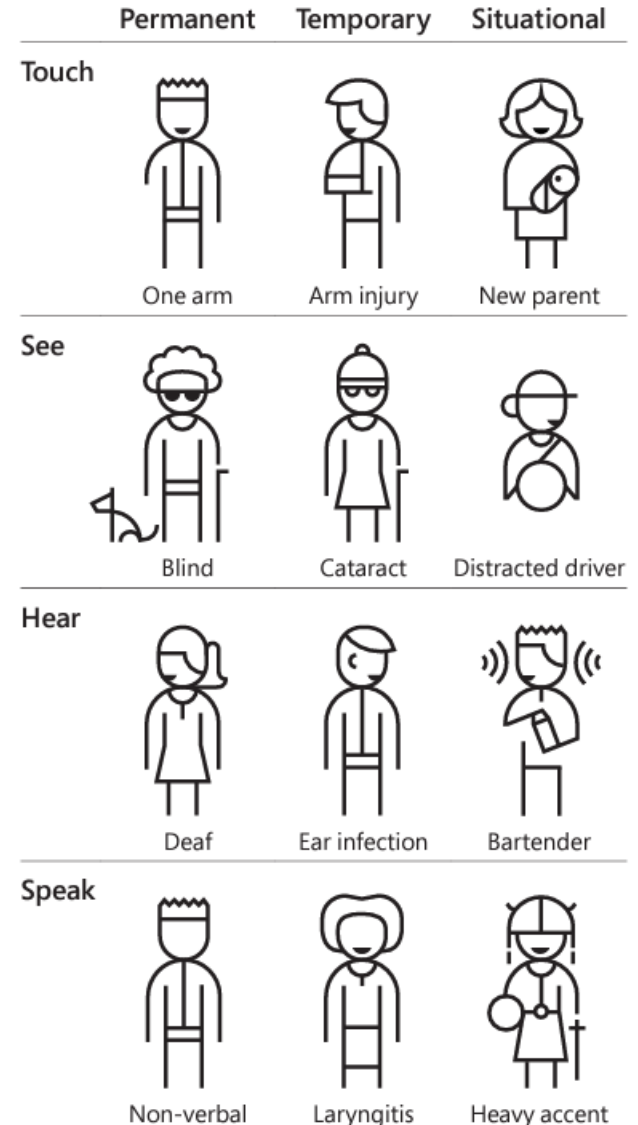


LinkedIn



The Persona Spectrum

- We use the Persona Spectrum to Understand related mismatches and motivations across a **spectrum of permanent, temporary, and situational** scenarios.
- It's a quick tool to help foster empathy and to show how a solution scales to a broader audience.



Tapping Performance when Walking

- Task: tap with stylus on targets of varying sizes and distances
- Conditions:
 - Sitting
 - Treadmill: slow
 - Treadmill: fast
 - Obstacle course (self-paced)
- Measures:
 - Pointing speed
 - Errors



Fig. 1. Demonstration of someone walking on the treadmill while doing target selection task.

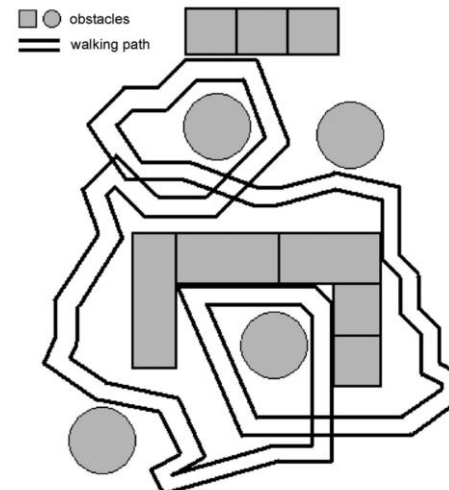
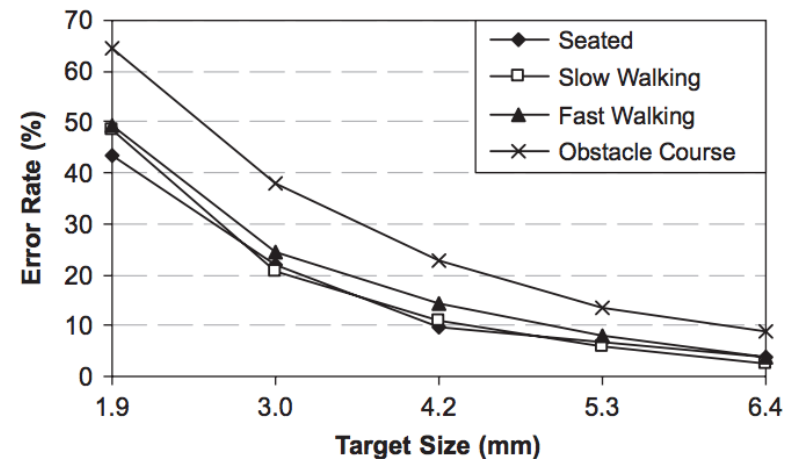
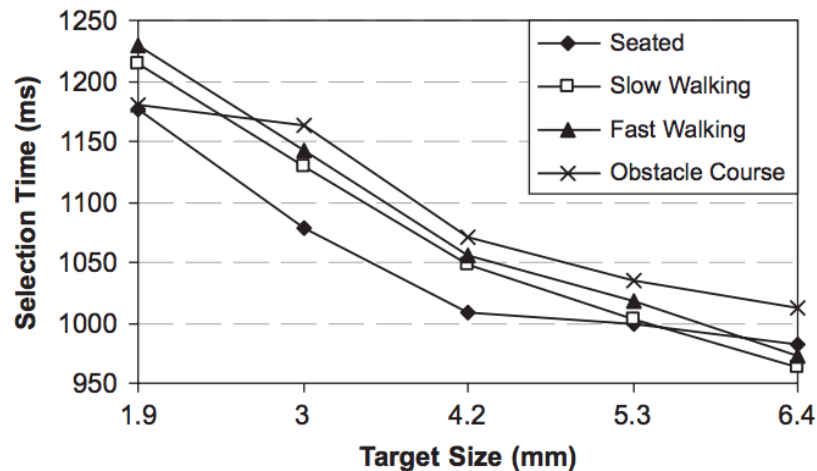


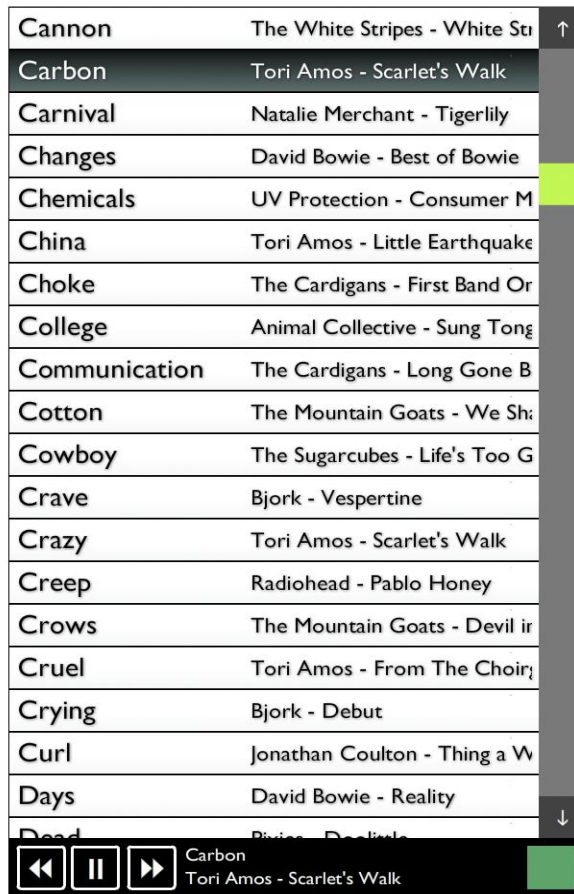
Fig. 2. Diagram of the obstacle course path used to create the attention-intensive walking condition.

Lin et al. How do people tap when walking? An empirical investigation of nomadic data entry. *International Journal of Human-Computer Studies* pp. 759-769

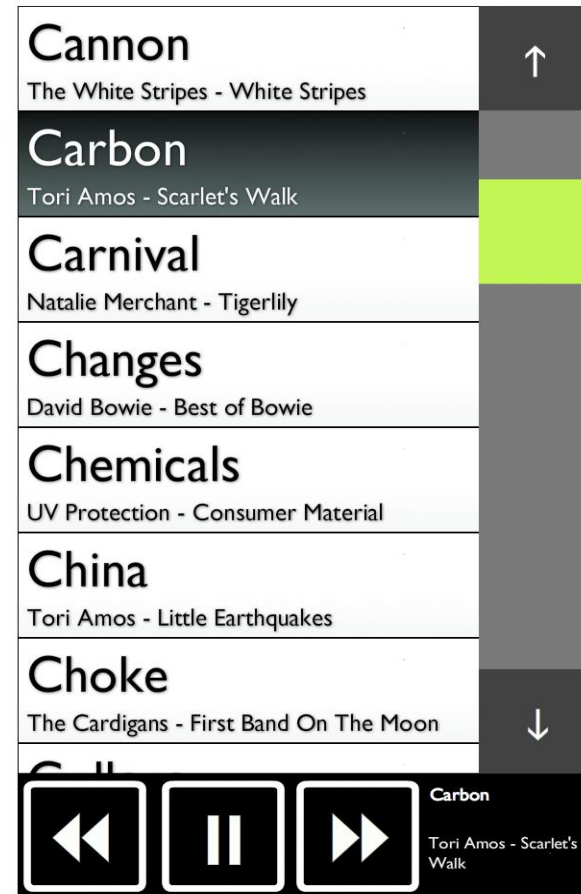
Tapping Performance when Walking

- The time to select individual targets did not differ significantly between conditions
- Error rates increased dramatically when the participants walked through the obstacle course even they reduced their walking speed by 36%.
- 90% or higher accuracy for targets with diameters as small as 4.2, 5.3, and 6.4 mm under the seated, treadmill, and the obstacle course conditions, respectively





Sitting UI



Walking UI

Kane et al. Getting off the treadmill: evaluating walking user interfaces for mobile devices in public spaces. MobileHCI '08: Proceedings of the 10th international conference on Human computer interaction with mobile devices and services (2008)

The Key Areas for Considering

- **Visual:** Long-sightedness, blindness, color blindness, are all forms of visual disability you need to cater for in your design.
- **Motor/Mobility:** This category doesn't just extend to problems with the use of the hands and arms, but also with other muscular or skeletal conditions.
- **Auditory:** Auditory disabilities affect the hearing and come in varying degrees of severity, up to and including total deafness.
- **Seizures:** Some individuals can be affected by light, motion, flickering, etc. on screen, thus triggering seizures.
- **Learning:** It's also important to remember that not all disabilities are physical. Learning and cognitive disabilities can also influence accessibility.

Some Suggestions

- If you use a CMS, choose one that **supports accessibility standards**. Drupal and WordPress, for example, support these
- Use **alt text** on your images; if you use images to enhance content, then a screen reader will need to explain them— that's what the alt text is for
- Have a **link strategy**. Instead of “Click here”. “Read more about the UBC, at their website.”
- Choose **colors carefully**; if in doubt, test your color schemes with some color-blind people
- **Don't refer just to the color**; “click the red button” might not be helpful to a color-blind person
- More information: <https://www.interaction-design.org/literature/article/accessibility-usability-for-all>

Curb Cut Phenomenon

- A design for people with impairments which benefits everyone
- Sidewalk curbs must be “cut” to allow wheelchair access
 - But benefits more than just wheelchair users
 - Easier to roll luggage, use strollers, carts, etc.
- Many technologies designed for those with disabilities have had this positive effect. More examples?



Midterm Review

What is HCI?

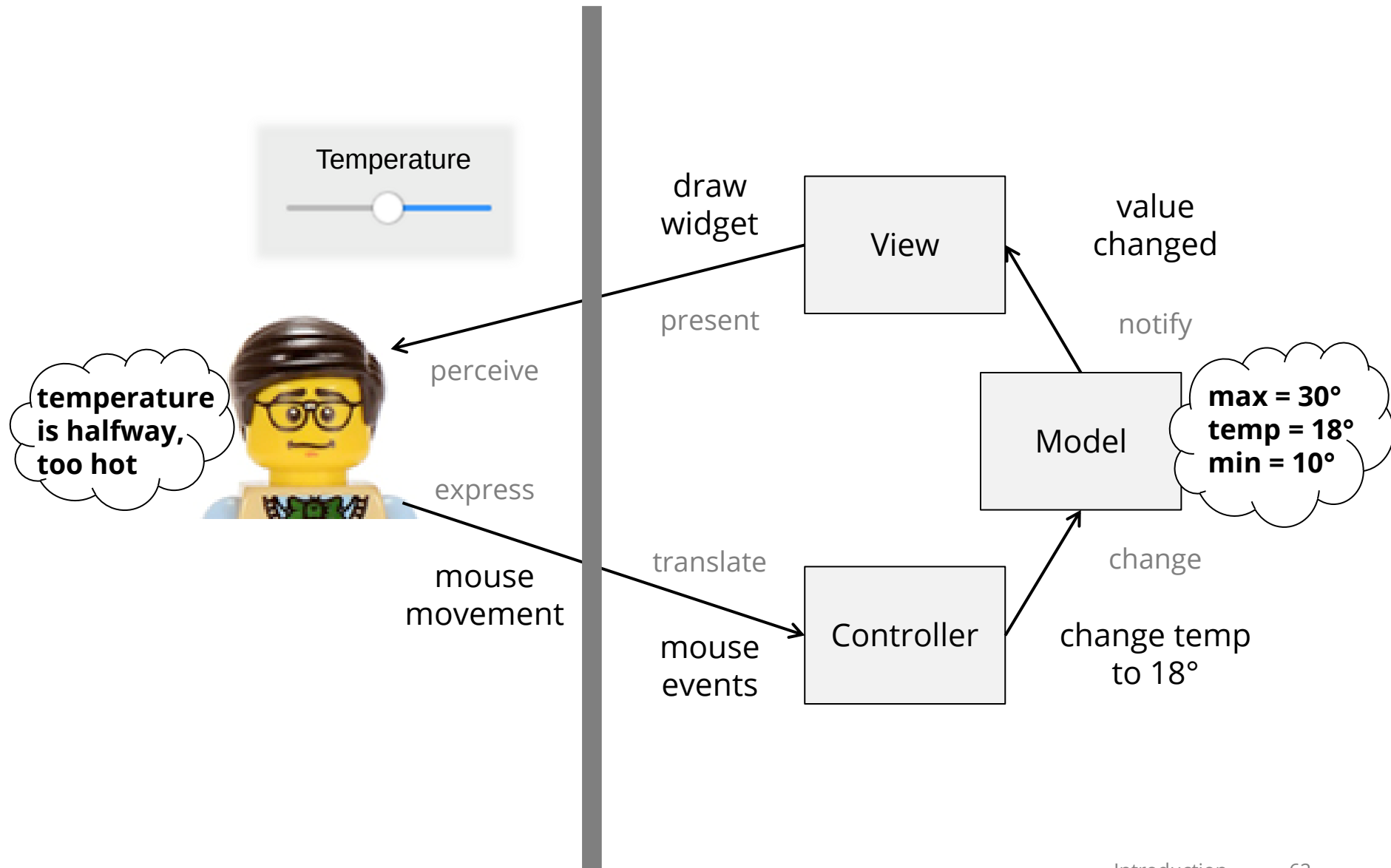
- “Human-computer interaction is a discipline concerned with the **design, evaluation and implementation of interactive computing systems** for **human use** and with the study of major phenomena surrounding them”.

-ACM definition

Interface vs. Interaction

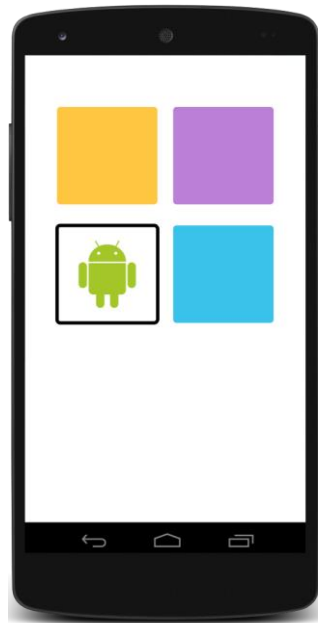
- What is the difference between an *interface* and *interaction*?
- **Interface** refers to the external presentation to the user
 - Controls (what you can manipulate to communicate intent)
 - Feedback (what the program uses to communicate its response)
- **Interaction** refers to actions by user and system over time
 - interaction is a dialog with a cycle alternating between the user manipulating controls and the system responding with feedback

Graphical Temperature Control

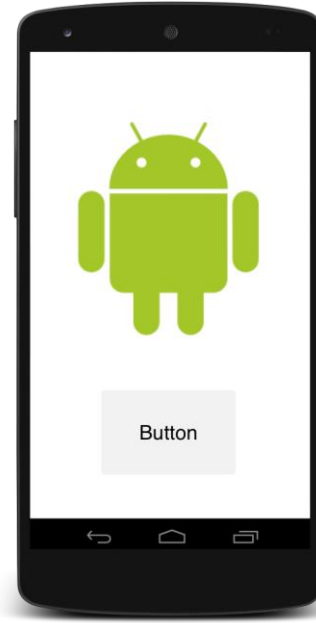


Activities

- A standard application component is an Activity
 - Typically represents a single screen
 - Main entry point (equivalent to main() method)
 - For most purposes, this is your application class



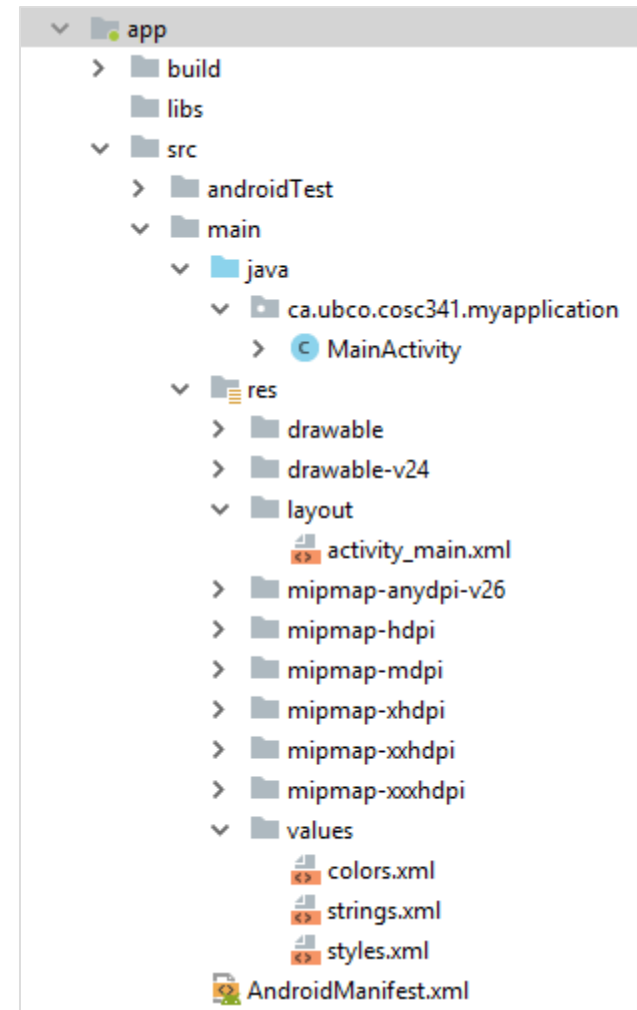
Activity 1

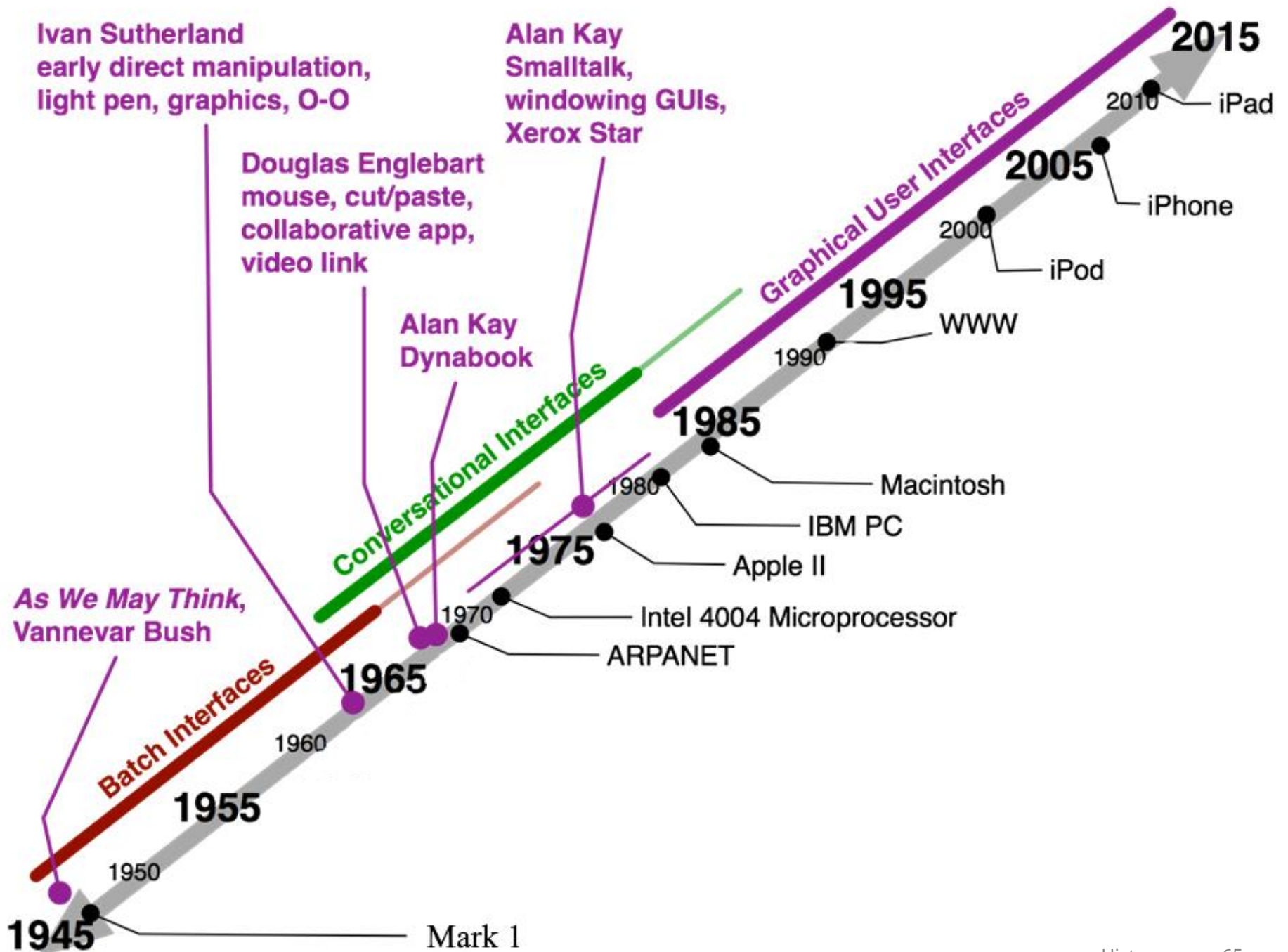


Activity 2

Android Project Files

- Manifest (app/manifests/)
 - Application setting
- Java (app/java/)
 - **(*.java) source code**
- Resources (app/res/)
 - **layout: (*.xml) UI layout and View definitions**
 - **values: (*.xml) constants** like strings, colours, ...
 - also bitmaps and SVG images (mipmap*, drawable*,)





Usability

- Usability is a quality attribute that assesses how easy user interfaces are to use
- Some usability goals include:
 - Effectiveness
 - Efficiency
 - Safety
 - Utility
 - Learnability
 - Memorability

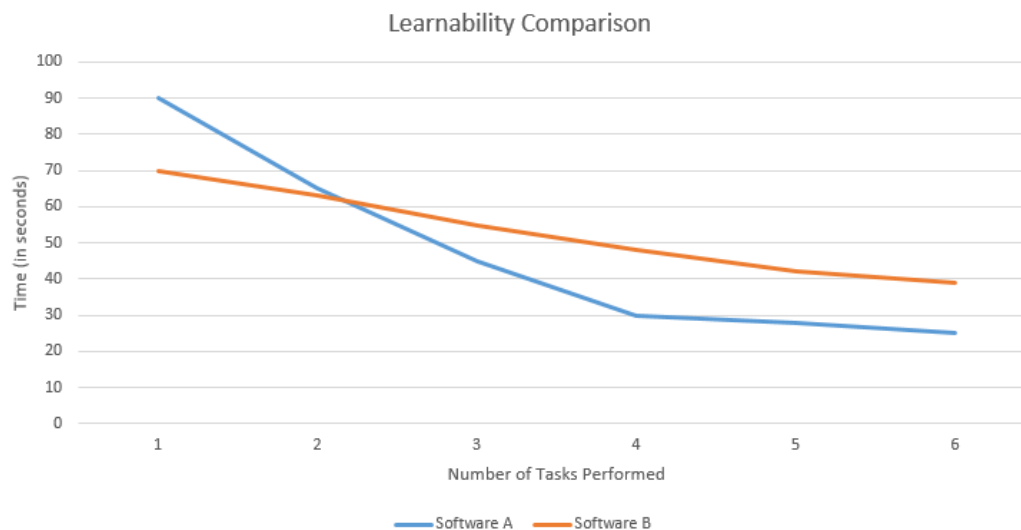
<https://www.nngroup.com/articles/usability-101-introduction-to-usability/>

Usability Goals

- Efficiency
 - Effective, ok; but efficient?
 - Considers how much time/number of steps it will take users to perform their tasks
 - Systems that require the least time/number of steps are usually considered as being efficient
 - Criteria: time to complete a task; # of operations to complete a task

Example Usability Goal

- Learnability
 - important if system will be adopted by the user
 - A more learnable system is one that reduces the time it takes to complete tasks as users spend more time with a system faster than others
 - Question: can primary (core) and secondary tasks be learned quickly and easily?
 - Criteria: time to learn a task, errors made in learning a task
e.g. smartphone?



User Experience Goals

- A set of user experience goals include:
 - Satisfying: productive
 - Motivating: did not feel like giving up
 - Enjoyable: no frustrations were encountered
 - Aesthetically pleasing: subjective
 - Fun: excited about using it again
 - Supportive of creativity: drawing tools
 - Entertaining: games
 - Rewarding: sense of productivity
 - Helpful: clueless but still made it through

Generic Design Principles

- Visibility
- Feedback
- Constraints
- Mapping
- Consistency
- Affordance
- Simplicity
- Matching
- Minimizing memory load
- Diagnose/Recover errors
- Control and freedom
- Flexibility
- Provide Help

Categories of users

- Three categories of users:
 - Primary:
Frequent hands-on (everyday)
 - Secondary:
Occasional or via someone else (getting reports)
 - Tertiary:
Affected by its introduction (owner)

User Profile

- Collection of attributes for describing a collection of users
 - i.e., age, disabilities, gender, culture, experience, plus any other factors relevant to a particular system
 - E.g., ages 18-35
- A user profile will help you to understand who you are building your product for
- Most systems will have a number of different profiles

Persona

- Personas
 - Personas are **not real people**, but they represent them through the design process
 - A number of **fictitious users** with specific values for all of the relevant attributes
 - Rich descriptions of typical users
 - Make profiles more life-like

Requirements

- Five primary types:
 - Functional
 - Data
 - Environmental / Context of Use
 - User
 - Usability

Data Gathering Involving Users

- Primary methods
 - Interviews
 - Questionnaires
 - Observations
 - Focus groups
- Advantages and disadvantages of each method

Interviews



Questionnaires



Observation



Focus Group



Overview of techniques

Technique	Purpose	Data type
Interviews	Exploring issues	Mostly Qualitative
Questionnaires	Answering specific questions	Qualitative and Quantitative
Observation	Understanding context of user activity	Qualitative
Focus Group	Collecting Multiple Viewpoints	Mostly Qualitative

What is a Prototype?

- In interaction design it can be (among other things):
 - a series of screen sketches, i.e. not necessarily related
 - a storyboard, i.e. a cartoon-like series of scenes
 - a Powerpoint slide show, e.g., simulation
 - a video simulating the use of a system
 - a cardboard mock-up
 - a piece of software with limited functionality written in a language
- Key point:
 - Goal is to NOT the artifact, but the feedback

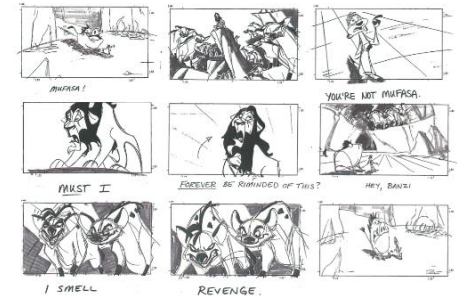
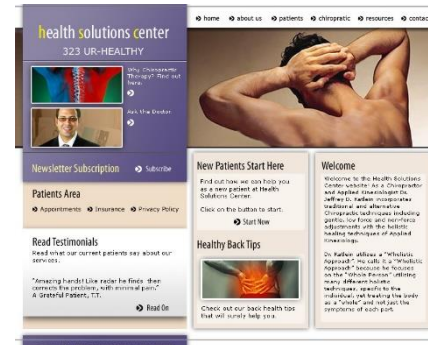
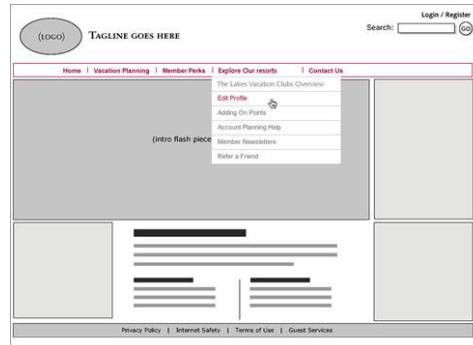
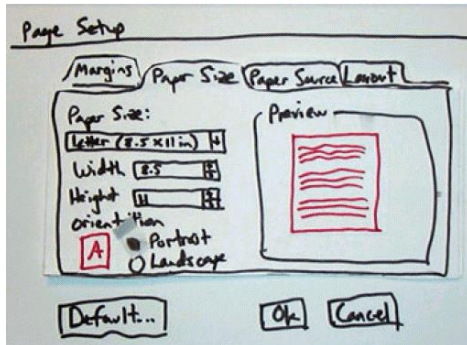


Image source: Nintendo

Prototype Fidelity

There are many types of prototypes, ranging anywhere between these two extremes:

- Low fidelity: prototype doesn't look much like final product, operation of the prototype may be simulated and slower.
- High fidelity: prototype looks and operates like the real product.



Why Prototype?

- **Communication:** discuss ideas with stakeholders
 - “Where’s the ON button?”
- **Evaluate** interface effectiveness for communicating conceptual model
 - “Whoops, users didn’t understand that.”
- Develop **requirements** and/or specifications •
 - “Uh-oh, here’s something we forgot.”
- **Learning** and problem solving
 - “Hey, that will work!”
- Further develop **conceptual and physical design**
 - “That’s way too heavy”
- **Save time and money**
 - Don’t waste time coding/building the wrong thing

What to Prototype?

- **Screen layouts and information display:** placement of different widgets, types of widgets, etc
 - e.g. should edit fields come after/before control buttons
- **Work flow, task design:** the sequence of steps required to achieve the task
 - e.g. flow from one screen to another, changes on the screen
- **Difficult, controversial, critical areas:** provide alternative solutions to be assessed
 - e.g. business critical reports with specific alignment of fields
- **Technical issues:** will the graphics rendered be fast enough?

When to prototype?

- To get out of an **early**, focused discussion, and to reach agreement
- when you need to **communicate ideas**
 - design team, managers, users etc.
- when you have questions and you can't proceed:
 - **functionality:**
structure, sequencing, flow, clarity & completeness of information
Example: knob versus slider for controlling volume
 - **appearance:**
branding, clarity, aesthetics, color, shape, etc.
screen too crowded? actions clear, or lost in clutter?
 - **specifications:**
“design by prototyping” (evolutionary approach)
how many files to show in file selection box

Prototypes

- Storyboarding/Paper prototype benefits
- Storyboarding/Paper prototype limitations
- What can/can't be learned from low/high-fidelity prototype
- Pros and cons of low/high-fidelity prototype
- Video prototype, wizard-of-oz

Limiting Functionality: How Much to Represent?

- Vertical prototypes
 - includes **in-depth functionality** for only a few selected features
 - common design ideas can be tested in depth
- Horizontal prototypes
 - surface layers includes the **entire user interface** with **limited underlying functionality**

